How much do you trust your data? Multiplicity-adjusted Bayesian hypothesis testing via the probability of direction

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Abstract

The probability of direction (pd) has seen rapid adoption in psychological research as an intuitive posterior summary that quantifies how strongly the posterior distribution favors one direction of an effect over the other. In practice, researchers often turn pd into a binary decision rule by applying fixed thresholds (e.g., $pd > 0.975$), declaring evidence for an effect when posterior mass concentrates strongly on one side. When this rule is applied across many parameters, it can substantially increase the risk of misleading conclusions. We argue that this problem stems from neglecting the joint prior probability that for every tested effect, the data-favored direction does not reflect a genuine effect. This quantity, often left implicit, becomes increasingly consequential as the number of tested effects grows. To address this, we propose a prior-odds adjustment to pd hat requires researchers to explicitly specify their belief about whether the effects favored by the data would be genuine. The adjustment accounts for multiplicity without modifying the model used in parameter estimation, offers a transparent mechanism for incorporating prior skepticism about the existence of effects, and reduces the risk of false discoveries in multi-parameter settings. We evaluate its behavior through simulation across varying sample sizes and proportions of null effects, and provide an R package to facilitate applied use.

Keywords: probability of direction, multiple testing, prior-odds

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Code is openly available in the *multibayes* repository on GitHub at https://github.com/mar-cald/multibayes. The authors declare that they have no competing interests.

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Over the past two decades, the use of Bayesian methods in psychological research has grown considerably (Jevremov & Pajić, 2024; Pfadt et al., 2025; Van De Schoot et al., 2017). This expansion has, however, been accompanied by practices reminiscent of the "null ritual" in frequentist statistics, where widespread adoption gave way to automatic application rather than principled reasoning (Cohen, 1994; Gigerenzer, 2004). For instance, researchers sometimes treat Bayes Factor thresholds (e.g., BF > 3) as dichotomous decision rules, or claim that an effect "exists" because the posterior probability of a directional effect exceeds a conventional benchmark such as 95% (Heck et al., 2023; Hoijtink et al., 2016; Kelter, 2023; Kruschke, 2018; Tendeiro et al., 2024; Tendeiro & Kiers, 2019). While such summaries are convenient, their use as default cutoffs encourages all-or-none interpretations. When these binary judgments are applied repeatedly across many parameters, they generate a multiplicity problem no different in kind from the one long recognized in frequentist statistics.

Yet even there, the problem of multiplicity has been handled inconsistently in practice. Researchers typically account for multiplicity in well-recognized cases, such as post hoc comparisons, but rarely do so when evaluating several effects simultaneously within the same model (Zeevi et al., 2025). This is routine in ANOVA or multiple regression, where main effects, interactions, and multiple predictors each constitute a distinct hypothesis test. Either way, the probability of at least one false positive accumulates with the number of tests performed, regardless of how those tests are framed or grouped (Bender & Lange, 2001; Cramer et al., 2016). Frequentist methods have been developed to address precisely this accumulation, targeting either the Family-Wise Error Rate (FWER; the probability of at least one false positive) or the False Discovery Rate (FDR; the expected proportion of false positives among rejected hypotheses) (Benjamini & Hochberg, 1995; Westfall & Young, 1993). These procedures address whether a spurious result enters the literature, but leave unresolved a subtler consequence of the same selection process. When researchers examine many parameters but highlight only the most extreme estimates, selection systematically inflates reported magnitudes (Altoè et al., 2020; Gelman & Carlin, 2014; Zwet & Cator, 2021).

Bayesian methods can mitigate this overestimation by pulling extreme estimates toward more plausible values, via informative priors (Dawid, 1994; Zwet & Gelman, 2022) and hierarchical modelling (Berry & Hochberg, 1999; Gelman et al., 2012, 2013; Kruschke, 2011). Crucially, however, magnitude bias and false positive accumulation are related but distinct problems. Shrinkage does not, on its own, provide any guarantee about the error properties of a repeated decision rule applied across many parameters: if the same cutoff is applied to many parameters (e.g., declaring an effect whenever the posterior probability exceeds 95%), the probability of spurious findings can remain substantial. Bayesian solutions to this issue typically rely on either simultaneous credible statements (Box & Tiao, 1992; Gelman & Tuerlinckx, 2000; Pruzek, 2016; Schnell et al., 2020) or priors over the hypothesis space that scale with the number of tests (Jeffreys, 1938; Scott & Berger, 2006; Scott & Berger, 2010; Westfall et al., 1997). Despite these available tools, hierarchical modeling alone is still routinely treated as sufficient for multiplicity control, a conflation that leaves the problem only partially addressed (Chen et al., 2020; Erčulj & Štrumbelj, 2015; Hogeveen et al., 2022; Limbachia et al., 2021; Mathar et al., 2022).

This issue becomes particularly salient when posterior summaries such as the probability of direction (pd) are used as decision rules. The pd quantifies the probability that an effect lies in its dominant direction (Makowski et al., 2019). In recent years, it has increasingly been adopted as a standardized decision criterion, with researchers claiming evidence for an effect when it exceeds fixed thresholds (e.g., $pd > 0.95$ or $pd > 0.975$) (Ben-Artzi et al., 2022; Bramley & Ruggeri, 2022; Brissenden et al., 2023, 2024; Ciranka et al., 2026;

Hochman et al., 2025; Makowski et al., 2020; Schurr et al., 2024; Stone-Sabali et al., 2024). Although pd is a fully Bayesian quantity, its use as a fixed decision rule connects naturally to familiar frequentist concerns: for location parameters in exponential family models under a uniform prior, pd equals the complement of the one-sided p-value, and this relationship holds approximately under a broader class of priors (Casella & Berger, 1987; Marsman & Wagenmakers, 2017). Because weakly informative or flat priors are commonly used in applied psychological research (Smid & Winter, 2020; Van De Schoot et al., 2017), a decision rule based on pd thresholds inherits the long-run error behavior of frequentist significance testing, including its susceptibility to multiplicity inflation. Yet this vulnerability has largely gone unrecognized.

To address this gap, we examine the frequentist operating characteristics[1] of thresholding rules of the form $pd > c$ and consider how the resulting multiplicity problem can be addressed. Specifically, we propose a prior-odds correction for pd grounded in the global null framework of Jeffreys (1938; see also Westfall et al., 1997) and Bayesian variable selection literature (Abraham et al., 2022; Efron et al., 2001; Guo & Heitjan, 2010; Scott & Berger, 2006; Scott & Berger, 2010). This approach requires specifying a prior probability that none of the claimed effects are real. It bears emphasis that the proposed adjustment does not control error rates per se; rather, it makes explicit the prior probability assigned to this event, allowing the resulting posterior odds to reflect both the observed data and the researcher's substantive beliefs about the prevalence of true effects. When those beliefs are well-calibrated to the data-generating process, error rates are naturally reduced as a consequence.

The paper proceeds as follows. We begin by introducing pd and examining its role as a test for existence, discussing how prior choice shapes its behavior and why more skeptical within-model priors alone cannot resolve the multiplicity problem. We then develop the prior-odds adjustment and evaluate its properties through simulation. A worked example using the accompanying R package (Calderan & Gambarota, 2026) follows, after which we conclude with a general discussion.

Probability of direction

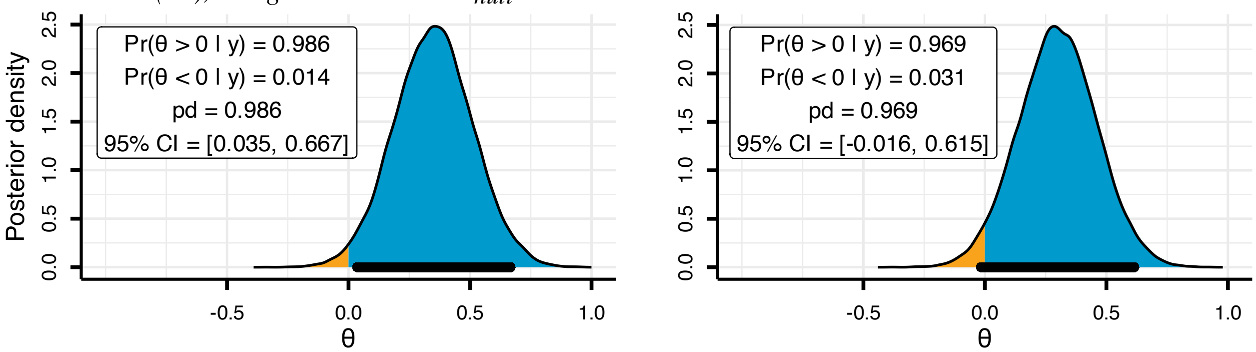
The probability of direction (pd) is a posterior summary that quantifies how strongly the posterior distribution favors one direction of an effect relative to a reference (θ_{null}) value (Makowski et al., 2019). For a continuous posterior, it can be defined as the larger of the two one-sided posterior probabilities: $pd = \max(\Pr(\theta < \theta_{null} \mid y),\ \Pr(\theta > \theta_{null} \mid y))$, and it ranges from 0.50 (no directional preference) to 1 (all on one side). Because pd is derived from the posterior, it depends on the full model specification (likelihood and prior), so the long-run error behavior of pd-based decisions is not a property of pd in isolation but of the overall Bayesian procedure that generates the posterior.

In practice, the pd is often used to summarize both the direction and existence of an effect. This interpretation is clearest when phrased as a decision rule $pd > c$: the posterior must assign more than c probability mass to the side of θ_{null} favored by the data. Setting $c = 1 - \alpha/2$ is convenient because it bounds the posterior mass in the opposite tail by $\alpha/2$. For concreteness, suppose the posterior favors a positive effect so that $pd = \Pr(\theta > 0 \mid y)$. An $\alpha = 0.05$ implies $c = 0.975$, therefore the rule $pd > c$ requires that $\Pr(\theta \leq 0 \mid y) < 0.025$, placing less than 2.5% of posterior mass at or below 0.

[1] Although frequentist and Bayesian frameworks rest on different foundations, examining the frequentist operating characteristics of a Bayesian procedure (how it behaves over repeated use) is a well-established practice (Müller et al., 2004; Ventz & Trippa, 2015; Zhang et al., 2019). This perspective treats long-run performance as a useful diagnostic for Bayesian inference (Gelman et al., 2013; Rubin, 1984)

Figure 1

Posterior distributions illustrating the relationship between the probability of direction (pd) and 95% Credible Interval (CI), using c = 0.975 and θ_{null} = 0.



As Figure 1 illustrates, observing $pd > 0.975$ is equivalent to excluding 0 from the central 95% equi-tailed credible interval. When the credible interval excludes the null value, one can infer that the null is relatively implausible (Box & Tiao, 1992). Given these considerations, although pd and the frequentist p-value target different quantities, it is expected that decision rules of the form $pd > c$ approximate the operating characteristics of familiar frequentist rules such as $p < \alpha$.

Although we express the decision threshold as $c = 1 - \alpha/2$ (with $\alpha = 0.05$) and set $\theta_{\text{null}} = 0$ throughout for ease of exposition, neither choice is restrictive. The threshold c can be set to any posterior probability cutoff reflecting the researcher's tolerance for uncertainty, and θ_{null} can take any non-zero value, which is particularly relevant when hypotheses involve a minimum effect of practical interest (Lakens et al., 2018).

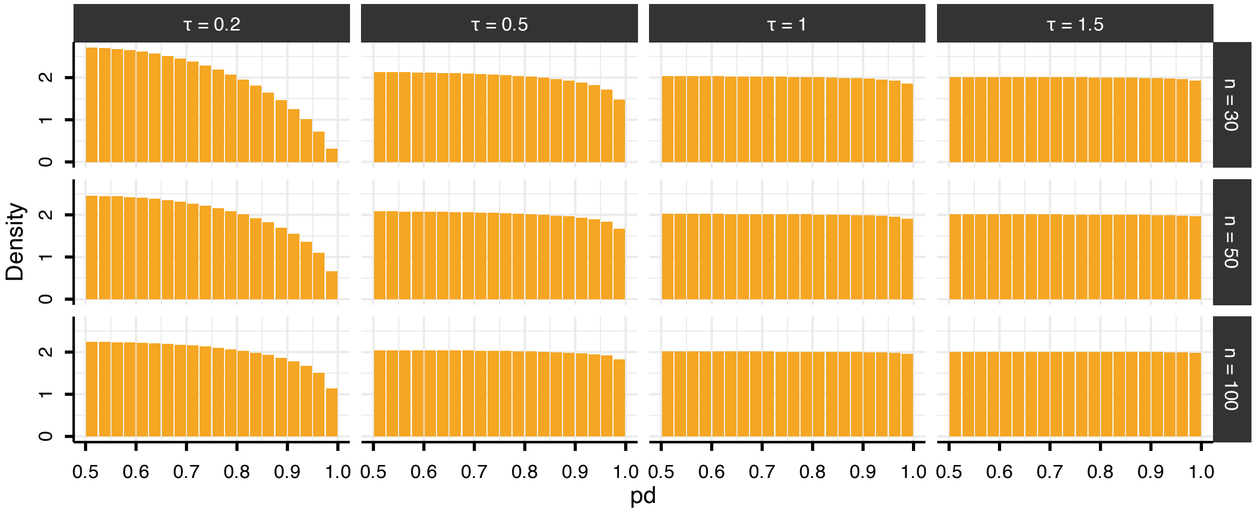
Multiplicity problem

Under the null hypothesis, p-values are uniformly distributed on $[0, 1]$. Because pd can be mapped to the p-value, this implies a corresponding null reference behavior for pd on $[0.5, 1]$. However, because pd is a posterior probability regarding the sign of θ, its null distribution is not determined by the likelihood alone: it can shift depending on how strongly the prior concentrates probability near 0 (i.e., how strongly the prior "expects" near-null effects). Consequently, even in the absence of multiple testing, the probability that pd exceeds 0.975 when the null hypothesis is true is not constant across priors and sample sizes (Figure 2). For a normal posterior[2], under diffuse priors ($\tau \geq 1$) and/or sufficiently large sample sizes, pd exhibits a distribution with constant density on the interval $[0.5, 1]$. Conversely, the use of a skeptical prior (e.g., $\tau = 0.2$) induces shrinkage toward 0.5, effectively reducing the error rate.

[2]To illustrate this, we use a conjugate normal-normal model with known variance, in which the posterior distribution and the null distribution of pd admit closed-form expressions. We varied sample size $n \in 30, 50, 100$ and prior scale $\tau \in 0.2, 0.5, 1, 1.5$, ranging from skeptical to relatively diffuse priors.

Figure 2

Distribution of the probability of direction (pd) stratified by sample size ($n = 30, 50, 100$; rows) and prior scale ($\theta \sim \mathcal{N}(0, \tau^2)$ with $\tau \in \{0.2, 0.5, 1, 1.5\}$; columns). Bin width = 0.025.



While skeptical priors reduce the probability that any single test yields an extreme pd, they do not address the issue that, as the number of tests m increases, so does the probability of observing at least one extreme value among $pd_1, \ldots, pd_m$. The approach presented below addresses this at the level of the testing procedure rather than the within-model prior, by introducing a separate prior on the joint event that all tested effects are simultaneously in the direction disfavored by the data. This allows researchers using uninformative or weakly informative priors to nonetheless incorporate substantive beliefs at the family level.

As we formalize in the next section, a specific condition must be met for pd to be used in this framework. Because pd represents the posterior probability mass on one side of θ_{null}, it maps directly onto the posterior odds between directional hypotheses ($\theta > \theta_{\text{null}}$ vs. $\theta < \theta_{\text{null}}$). These posterior odds equal the Bayes Factor only when the within-model prior is symmetric around θ_{null}, ensuring that each direction is equally plausible a priori, under which the prior odds are unity and the posterior odds reduce to the ratio of marginal likelihoods. This requirement is typically satisfied by the default priors in standard Bayesian software and remains compatible with any degree of prior skepticism regarding effect magnitude (i.e., via θ_{null}).

Prior-Odds adjustment for multiplicity

From posterior probabilities to posterior odds

To connect pd to posterior odds, consider the directional hypotheses $H_+ : \theta > 0$ and $H_- : \theta < 0$. These hypotheses partition the parameter space based on the sign of θ. If the prior for θ is continuous and symmetric about 0, then the two directions are equally plausible a priori: $\Pr(H_+) = \Pr(H_-) = 0.5$.

Without loss of generality, assume the posterior favors H_+, such that $pd = \Pr(H_+ \mid y) = \Pr(\theta > 0 \mid y)$. The complement is then given by $1 - pd = \Pr(\theta \leq 0 \mid y)$. The posterior odds comparing these two hypotheses are therefore:

$$\frac{pd}{1 - pd} = \frac{\Pr(H_+ \mid y)}{\Pr(H_{-0} \mid y)}$$

For a continuous posterior distribution, the probability of any single value (e.g., $\Pr(\theta = 0 \mid y)$) is 0, which means that one could write $\frac{pd}{1-pd} = \frac{\Pr(H_+|y)}{\Pr(H_-|y)}$. However, we retain the notation H_{-0} in the denominator to explicitly indicate that the hypothesis includes the point null value of zero. Applying Bayes' rule, these posterior odds can be decomposed into a Bayes Factor and prior odds:

$$\frac{\Pr(H_+ \mid y)}{\Pr(H_{-0} \mid y)} = \mathrm{BF}_{+/-0} \times \frac{\Pr(H_+)}{\Pr(H_{-0})}$$

where $\mathrm{BF}_{+/-0}$ quantifies the relative likelihood of the data under H_+ versus H_{-0}. Under prior symmetry, $\frac{\Pr(H_+)}{\Pr(H_{-0})} = 1$, which yields $\frac{pd}{1-pd} = \mathrm{BF}_{+/-0}$ (Casella & Berger, 1987; Marsman & Wagenmakers, 2017). Thus, the odds $\frac{pd}{1-pd}$ computed from posterior draws can be interpreted directly as evidence favoring $\theta > 0$ over $\theta \leq 0$ (or $\theta < 0$ over $\theta \geq 0$).

Multiplicity adjustment via prior-odds

Assessing a single directional claim under a symmetric prior centered at θ_{null} corresponds to prior indifference between the two hypotheses: $\Pr(H_+) = \Pr(H_{-0}) = 0.5$. When many such claims are considered jointly, however, retaining a prior probability of 0.5 for each implicitly assigns near-certainty to the event that at least one effect lies in its claimed direction. A principled response is to redistribute prior probability toward the data-disfavored hypothesis for each individual claim, as a function of the family size m (Jeffreys, 1938; Westfall et al., 1997).

Formally, for each test $i = 1, \ldots, m$, let $H_{1,i}$ denote the directional claim corresponding to the posterior-dominant sign used to compute $pd_i = \Pr(H_{1,i} \mid y)$, and let $\Pr(H_{0,i} \mid y)$ be its complement. The global null hypothesis, $\mathbf{H}_0$, is the event that, for every parameter, the data-favored direction does not reflect a genuine effect — equivalently, that no tested effect lies in its data-favored direction. Under exchangeability and independence, the prior probability of $\mathbf{H}_0$ can be defined as $\Pi_0 = \pi_0^m$, so specifying Π_0 implies a per-test prior probability

$$\pi_0 = \Pi_0^{1/m}.$$

Thus, π_0 is the marginal prior probability that a given effect is absent. Under the same assumptions, the number of such cases follows $\text{Binomial}(m, \pi_0)$, so π_0 corresponds to the expected proportion within the family. With π_0 in hand, the multiplicity adjustment enters entirely through the prior odds. For each test, the adjusted posterior odds satisfy:

$$\frac{pd_{\text{adj},i}}{1 - pd_{\text{adj},i}} = \frac{pd_i}{1 - pd_i} \times \frac{1 - \pi_0}{\pi_0}$$

which can be expressed as the adjusted posterior probability:

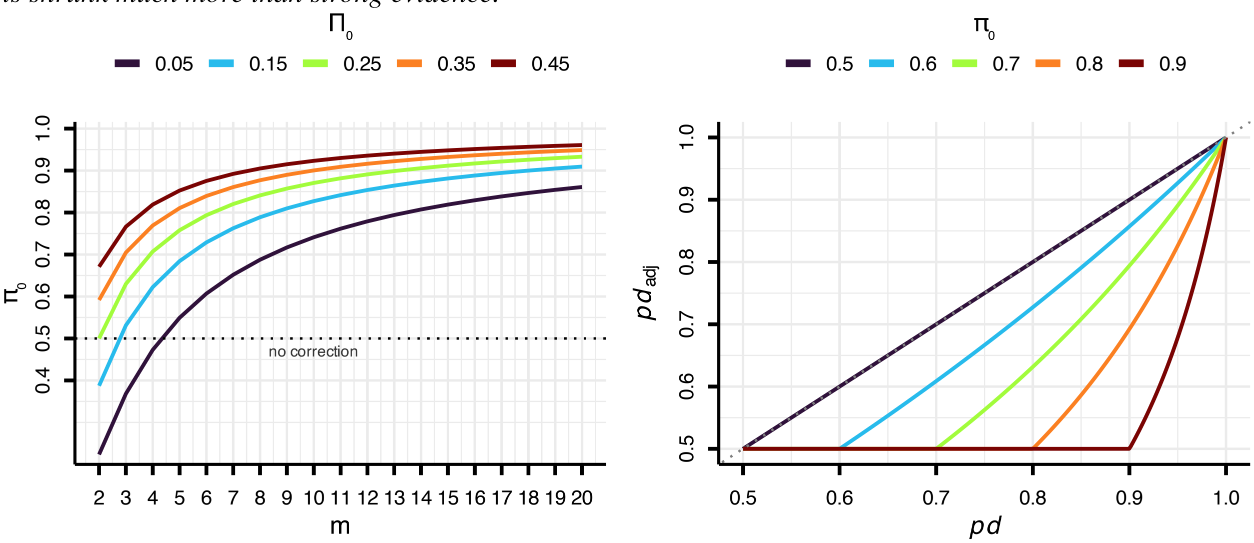
$$pd_{\text{adj},i} = \frac{pd_i \cdot (1 - \pi_0)}{pd_i \cdot (1 - \pi_0) + (1 - pd_i) \cdot \pi_0}.$$

Because the prior is symmetric and pd_i is always the posterior probability of the dominant direction, the factor $(1 - \pi_0)/\pi_0$ acts as a single scalar penalty regardless of the effect's sign.

Researchers encode their expectations through the choice of Π_0: lower values are appropriate when effects are well established, whereas higher values reflect greater uncertainty, as in exploratory research. For a given Π_0, larger families lead to higher π_0, but, unlike classical uniform threshold adjustments, the correction acts nonlinearly on posterior evidence, with higher pd values less affected by the correction (see Figure 3).

Figure 3

The left panel shows that the implied prior probability π_0 assigned to each hypothesis $H_{0,i}$ increases with family size m, depending on the specified global null probability Π_0. The dotted line marks the boundary condition $\pi_0 = 0.5$, below which the prior would favor H_1 a priori. The right panel shows the corresponding prior-odds adjustment of pd for different π_0 values, highlighting the heterogeneous penalty: weak evidence is shrunk much more than strong evidence.



A boundary condition arises when m is small and Π_0 is very low, in which case the formula can yield $\pi_0 < 0.5$, inflating the posterior odds in favor of H_1 rather than penalizing them. We therefore cap π_0 at 0.5, ensuring the adjustment acts exclusively as a conservative safeguard; when no penalty is warranted, $pd_{\text{adj},i} = pd_i$. Furthermore, since $pd \in [0.5, 1]$, the adjusted probability of direction has also bounded below by 0.5.

Simulation Study

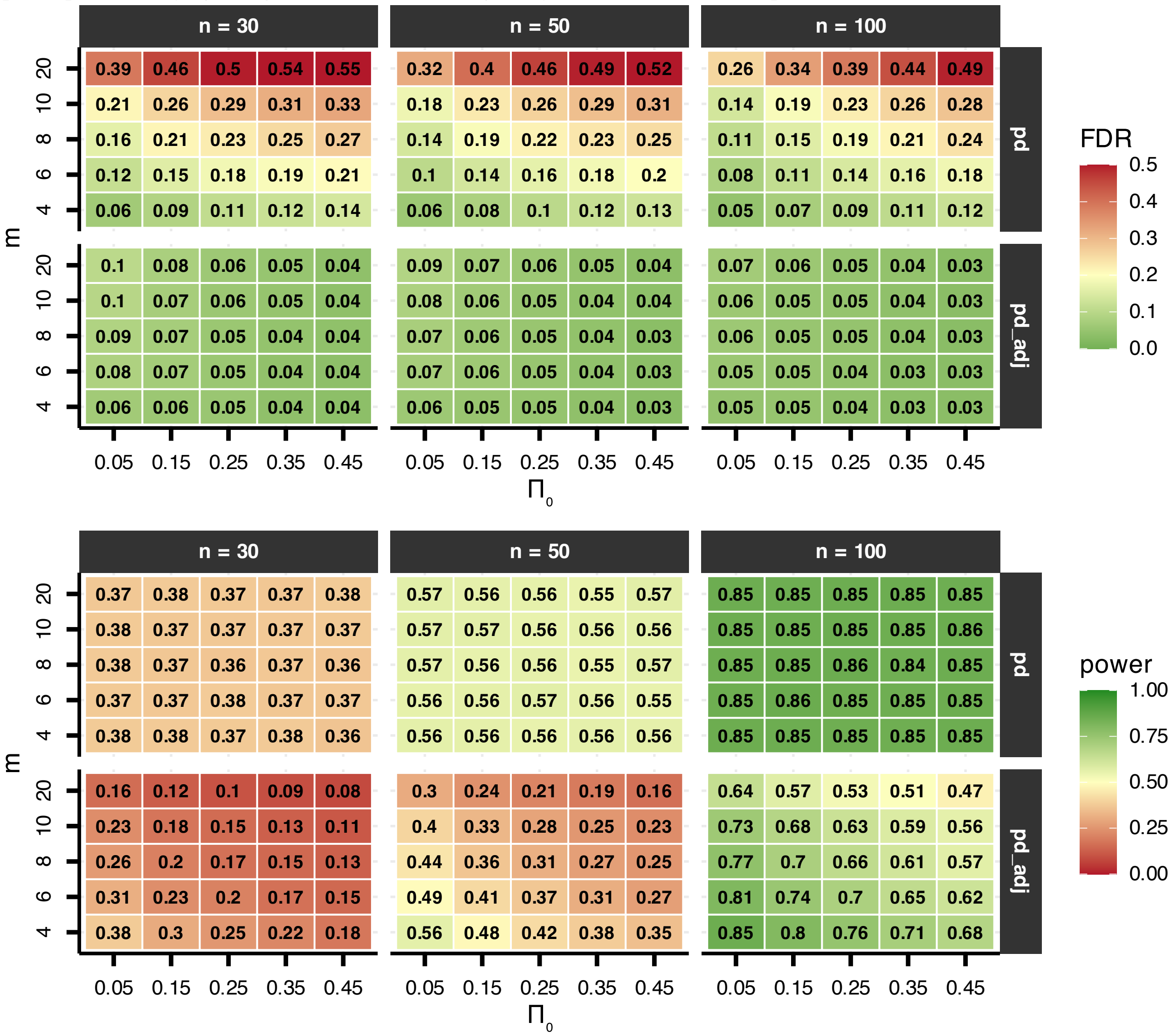
To examine the impact of Π_0, we conducted a simulation study with a fully crossed factorial design, varying family size $m \in \{4, 6, 8, 10, 20\}$, sample size $n \in \{30, 50, 100\}$, and the researcher-specified global null prior $\Pi_0 \in \{0.05, 0.15, 0.25, 0.35, 0.45\}$, with 10,000 replications per condition. In this first study, Π_0 was set to match the true data-generating null proportion $\pi_{0_{true}}$ $(= \Pi_0^{1/m})$: at each replication, the number of true null effects ($\theta_i = 0$) was drawn from a Binomial$(m, \pi_{0_{true}})$, and the remaining were assigned a non-null effect of $\theta_i = 0.3$. Data were drawn from a standard normal distribution, $y_i \sim \mathcal{N}(\theta_i, 1)$. For every simulated dataset, we computed the posterior distribution of θ using a conjugate normal-normal model with $y_i \mid \theta \sim \mathcal{N}(\theta, 1)$ and prior $\theta \sim \mathcal{N}(0, 2)$.

We considered the alternative hypothesis supported when $pd > 0.975$. Performance was evaluated in terms of the False Discovery Rate (FDR), defined as the proportion of false positives among all tests for which the null was rejected, and average power, defined as the proportion of true effects correctly detected (Benjamini & Hochberg, 1995; Izmirlian et al., 2025).

As shown in Figure 4, unadjusted pd-based decisions produced substantial FDR inflation, with error rates increasing as both Π_0 and family size m increased. In contrast, the adjustment procedure performed as intended. In this simulation, $\pi_{0\text{true}} = \pi_0 = \Pi_0^{1/m}$, meaning the specified prior was correctly calibrated; because Π_0 is explicitly incorporated into the correction, FDR did not inflate as a function of m, with values reaching a maximum of 0.10 in small samples. As expected, power decreased as a function of n, Π_0, and m, reflecting the more stringent penalty imposed under smaller samples, larger families, and higher null probabilities.

Figure 4

False Discovery Rate (FDR) and power of the unadjusted and adjusted pd procedure as a function of the prior probability of the global null (Π_0), the family size m, and the sample size n.



To examine the consequences of prior misspecification, rather than varying Π_0 directly, we parametrized the simulation in terms of the implied per-test null probability $\pi_0 \in \{0.50, 0.55, \ldots, 0.95\}$, from which the family-level prior is recovered as $\Pi_0 = \pi_0^m$. The true proportion of null hypotheses $\pi_{0\text{true}} \in \{0.5, 0.6, \ldots, 1.0\}$, family size $m \in \{4, 8, 10, 20\}$, and sample size $n \in \{30, 50, 100\}$ were varied independently, with each condition replicated 10,000 times.

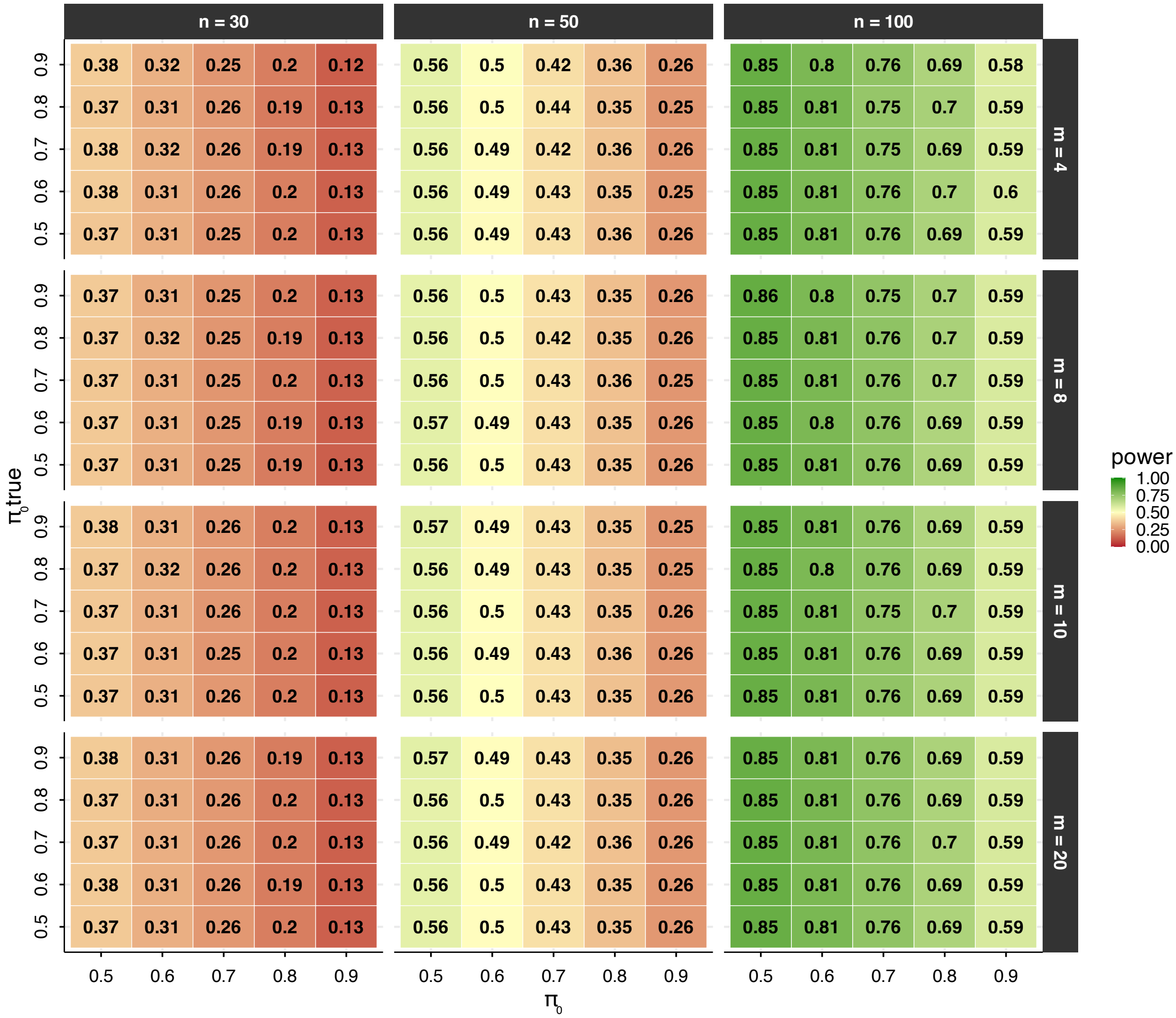
As shown in Figure 5, FDR remains near the nominal level across all conditions when m is small ($m = 4$), regardless of sample size, and proves robust even under mild misspecification ($\pi_0 \neq \pi_{0\text{true}}$). A key limitation arises when $\pi_0 = 0.5$ and $m > 4$, for which $\Pi_0 \to 0$; in small samples, noise can then lead to upward bias in FDR even when $\pi_0 = \pi_{0\text{true}}$. However, larger samples mitigate this issue, with FDR remaining low along the diagonal ($\pi_0 = \pi_{0\text{true}}$). As m increases, FDR rises substantially, particularly when $\pi_{0\text{true}}$ is large and the specified π_0 is small, that is, when the prior assumption underestimates the true proportion of null hypotheses. Consistent with the previous results, the correction reduces average power, with power decreasing as π_0 increases (Figure 6).

Figure 5

False Discovery Rate (FDR) of the adjusted pd procedure as a function of the prior probability assigned to the null for each test, π_0, the family size m, and the sample size n. The y-axis represents $\pi_{0_{true}}$, the true probability that each individual hypothesis is null. Each cell shows the empirical FDR averaged over 10,000 replications.

Figure 6

Average power of the adjusted pd procedure as a function of the prior probability assigned to the null for each test, π_0, the family size m, and the sample size n. The y-axis represents $\pi_{0_{true}}$, the true probability that each individual hypothesis is null. Each cell shows the empirical power averaged over 10,000 replications.



Application: worked example with the R Package `multibayes`

The procedure is implemented in the `multibayes` R package, which provides functions for computing adjusted pd values from either raw posterior draws or a vector of unadjusted pd values. The core function, `pd.adjust()`, accepts posterior draws directly and computes pd values internally when needed. The output includes the adjusted pd values, the tested null value, the implied global null probability, the number of tests m, and the implied per-test null probability. Unadjusted pd values are retained alongside adjusted ones to facilitate direct comparison; when posterior draws are provided as input, the posterior mean for each parameter is also returned.

For this example, we use the openly available dataset from Lin et al. (2020), who investigated the effect of ego depletion on conflict resolution across four studies. In their paradigm, an initial high-demand (vs. low-demand) task was followed by a Stroop task. For simplicity, we focus on Study 1, which tested the effects of congruency and ego depletion (within subjects). The dataset is readily accessible via the `afex` package (Singmann et al., 2016) and was chosen for this reason; this example is not intended as a critique of the original work. This dataset includes data from 253 participants; for this example, we randomly selected 80 participants.

Given the repeated-measures structure of the data and the large number of trials per participant, we fit a multilevel generalized linear model using `brms` (Bürkner, 2017) on reaction time (RT) data. A Lognormal model was specified, with default flat priors; these were adopted not as a principled recommendation, but to illustrate the method's behavior under the most common applied scenario, which also represents the condition where the prior-odds adjustment is most suggested.

The model include the effect of congruency (congruent vs. incongruent), previous congruency (congruent vs. incongruent), condition (depletion vs. control), and trial number (scaled within participant and condition). The sequential congruency effect was modeled as the interaction between congruency and previous congruency; the hypothesis that depletion amplifies the congruency effect over time was captured by the three-way congruency × condition × trial interaction. This yielded nine hypotheses (Table 1)[3].

Although the theoretical hypotheses are directional (e.g., incongruent trials are expected to slow responses), pd is computed uniformly as $\max\big(\Pr(\hat{\theta} > \theta_{\text{null}}),\ \Pr(\hat{\theta} < \theta_{\text{null}})\big)$ for all hypotheses. This direction-agnostic formulation tests whether the null value falls inside or outside the posterior credible region, without presupposing a predicted direction. Note that hypotheses are tested on the log scale of the posterior. Non-zero null values were specified for all main effects besides `condition`, and for the sequential congruency effect.

Table 1

Tested hypotheses for RT, evaluated on the log scale of the posterior.

Hypothesis	**Null value**
H_1: congruency effect	0.05
H_2: previous congruency effect	0.05
H_3: condition effect	0
H_4: trial number effect	0.05
H_5: congruency × previous congruency interaction	0.05
H_6: congruency × condition interaction	0
H_7: congruency × trial interaction	0
H_8: trial × condition interaction	0
H_9: congruency × condition × trial interaction	0

We retained only correct trials preceded by a correct trial (i.e., error and post-error trials were excluded), yielding $5,965$ observations.

```
data_rt <- data_stroop |>
  group_by(id) |>
  filter(acc == 1 & lag(acc) == 1) |>
  ungroup()

# set contrast
```

[3]In this example we included all nine tested hypotheses in the family, although one could reasonably argue that not all of them bear on the question of whether ego depletion affects conflict resolution (e.g., the main effect of trial number). In such cases, researchers should clearly justify their choice of family and the rationale for including or excluding specific hypotheses.

```
contrasts(data_rt$congruency) <- contr.sum(2)/2
contrasts(data_rt$prev_cong) <- contr.sum(2)/2
contrasts(data_rt$condition) <- contr.sum(2)/2

mod_rt <- brm(formula = rt ~ 1 + congruency * prev_cong +
  congruency * trial * condition +
  (1 + congruency + prev_cong +
     congruency + trial + condition || id),
              family = lognormal(), data = data_rt,
              cores = 3, chains = 3, iter = 8000)
```

Since `pd.adjust()` computes pd values automatically from a matrix of posterior draws, we extracted the relevant draws from the model.

```
draws_rt <- as.data.frame(mod_rt)
par_names <- colnames(draws_rt)[2:10]
draws_rt  <- draws_rt[, par_names]
pd.adjust(draws = draws_rt, p = 0.02,
          null.value = c(0.05, 0.05, 0.05, 0, 0.05, 0, 0, 0, 0))
```

```
                               mean.est null.value     pd pd.adj    p m     pm
b_congruency1                   -0.1440       0.05 1.0000 1.0000 0.02 9 0.6475
b_prev_cong1                    -0.0166       0.05 1.0000 1.0000 0.02 9 0.6475
b_trial                          0.0119       0.05 1.0000 1.0000 0.02 9 0.6475
b_condition1                     0.0064       0.00 0.6509 0.5038 0.02 9 0.6475
b_congruency1:prev_cong1         0.0222       0.05 0.9866 0.9756 0.02 9 0.6475
b_congruency1:trial             -0.0033       0.00 0.7206 0.5840 0.02 9 0.6475
b_congruency1:condition1        -0.0165       0.00 0.9291 0.8770 0.02 9 0.6475
b_trial:condition1               0.0071       0.00 0.8936 0.8205 0.02 9 0.6475
b_congruency1:trial:condition1  -0.0028       0.00 0.5961 0.5000 0.02 9 0.6475
                               direction
b_congruency1                  two.sided
b_prev_cong1                   two.sided
b_trial                        two.sided
b_condition1                   two.sided
b_congruency1:prev_cong1       two.sided
b_congruency1:trial            two.sided
b_congruency1:condition1       two.sided
b_trial:condition1             two.sided
b_congruency1:trial:condition1 two.sided
```

The function was called with $\Pi_0 = p = 0.02$, yielding a per-test prior probability $\pi_0 = pm = p^{1/m} \approx 0.65$. Inspecting the output, three parameters — `b_congruency1`, `b_prev_cong1`, and `b_trial` — show $pd = 1.00$ both before and after adjustment, indicating strong evidence for the alternative hypotheses that is unaffected by the multiplicity correction. The interaction term `b_congruency1:prev_cong1` shows a modest reduction from $pd = 0.987$ to $pd_{\text{adj}} = 0.976$. In contrast, parameters with weaker evidence, such as `b_condition1` ($pd = 0.651$), `b_congruency1:trial` ($pd = 0.721$), and the three-way interaction `b_congruency1:trial:condition1` ($pd = 0.596$), are shrunk substantially toward 0.5 after adjustment.

Discussion

Thresholding pd into a decision rule and applying it repeatedly across many parameters induces a multiplicity problem. Without adjustment, researchers implicitly accept a prior probability for the global null hypothesis that may be far lower than intended, a concern particularly acute when non-informative or weakly informative priors are used for estimation. The approach developed here addresses this by recasting multiplicity as a problem of prior specification: building on the global null formulation of Jeffreys (1938; see also Westfall et al., 1997), the correction requires researchers to explicitly assign a prior probability to the global null event. The implied Bayes Factor $pd/(1pd)$ compares two directional hypotheses whose assignment is determined post hoc by the posterior, with the numerator defined as the data-favored direction. Under within-model prior symmetry (where priors are centered at θ_{null}) each such data-selected hypothesis is evaluated under prior odds of one, so that the procedure attributes equal a priori plausibility to the direction selected after inspection of the data. Across m tests, assuming independence, this is equivalent to assigning prior probability 0.5^m to the joint event that none of the data-favored directions corresponds to a genuine effect, a value that rapidly approaches zero as m grows. Making the prior on the global null explicit therefore serves two purposes. For researchers using diffuse or default within-model priors, it provides a principled way to encode skepticism at the family level without committing to informative priors on individual parameters. For researchers who would judge the implicit prior 0.5^m to be appropriate and decline the adjustment, it nonetheless renders that commitment transparent rather than implicit. In both cases, the procedure constitutes a transparent mechanism for encoding prior skepticism about data-favored hypotheses under multiplicity, and underscores the importance of carefully reflecting on the prior probability of the tested hypotheses.

Related corrections have been implemented for specific testing contexts, such as post-hoc tests in JASP (Jong, 2019; van den Bergh et al., 2020). The present procedure extends this logic to pd and provides a general-purpose implementation for applied use, applicable to any posterior-derived pd or directly to posterior draws, including cases where pd values arise from multiple separate analyses within the same manuscript (Blondé et al., 2021; Ciranka et al., 2026). This raises the question of how to define the hypothesis family to which the correction applies (Hochberg & Tamhane, 1987; Shaffer, 1995). Although no universal consensus exists regarding what constitutes a family, we favor the criteria proposed by Westfall et al. (2011), where a family is a set of questions that *"form a natural and coherent unit, [are] considered simultaneously in the decision-making process"*. Regardless of the chosen definition, authors should clearly specify their chosen family and explain how it serves the study's substantive goals. We further recommend that researchers justify their choice of prior probability for the global null by drawing on theory, expert elicitation, or domain expertise (O'Hagan, 2019). More generally, a high prior probability suits exploratory settings with many uncertain hypotheses, while a lower value is warranted in confirmatory settings. For applied reporting, the unadjusted pd should accompany one multiplicity-adjusted measure, with the number of tests and the implied global-null prior clearly stated.

Two limitations of the present framework warrant attention. First, the procedure assumes that all hypotheses are exchangeable and independent, assigning each the same prior probability of being null and ignoring dependencies among them. In practice, hypotheses are often dependent, for example when multiple tests rely on overlapping data. Bayesian approaches that explicitly model such dependence or adopt more flexible multiplicity structures have been developed (Gopalan & Berry, 1998; Guo & Heitjan, 2010; Malsiner-Walli & Wagner, 2016; Scott & Berger, 2006; Scott & Berger, 2010; Williams, n.d.), but their implementation requires substantially more modeling effort, limiting their accessibility in applied research.

Second, it does not provide evidence for a point null hypothesis. The procedure evaluates directional claims defined by inequalities (e.g., $\theta > \theta_{\text{null}}$ vs. $\theta \leq \theta_{\text{null}}$), and while the complementary hypothesis includes the null value, that value receives no special status. When the null falls outside the credible interval, one can infer its relative implausibility, but this is fundamentally different from quantifying evidence in its favor (Heck et al., 2023; Ly et al., 2016; Morey & Rouder, 2011). It is worth noting, however, that the prior-odds correction introduced here is not conceptually novel and can be extended to other Bayes Factor approaches. Rather than a method-specific innovation, it reflects a general principle of hypothesis testing: prior probabilities should be combined with all relevant evidence (Dienes, 2016). When joint modeling is feasible, a natural alternative is to integrate evidence across hypotheses via joint posterior probabilities (Box & Tiao, 1992; Gelman & Tuerlinckx, 2000; Pruzek, 2016; Schnell et al., 2020).

As Bayesian methods become increasingly common in psychological research, often in pragmatic workflows relying on default priors and fixed decision thresholds, the implications of multiplicity should be explicitly recognized. Additionally, although priors may be well-calibrated at the level of individual parameters, they do not by themselves account for multiplicity when many parameters are considered simultaneously. Ignoring this aspect leads fixed *pd* thresholds to systematically overstate evidence for directional effects, especially as the number of tests grows. The adjustment proposed here addresses this gap by incorporating prior beliefs about the prevalence of null effects directly into *pd*-based decisions. By doing so, it provides a principled and transparent way to account for multiplicity without altering the estimation model, yielding more reliable inferences in multi-parameter settings and reducing the risk that apparent effects reflect chance rather than substantive signals.

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